

Cybersecurity Study Notes

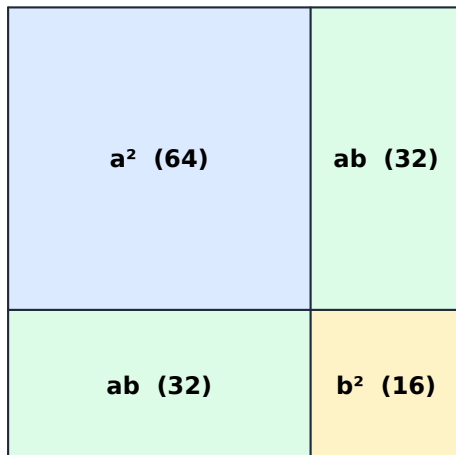
Cryptography basics, the CIA triad, and an SSH key-setup lab on AWS EC2

1. Where is $(a+b)^2$ used in cybersecurity?

In cybersecurity, $(a+b)^2$ is not a cybersecurity-specific formula. It is a mathematical concept that supports areas such as cryptography, algorithms, and security-related programming.

$$(a + b)^2 = a^2 + 2ab + b^2$$

Example with $a = 8$ and $b = 4$: $12^2 = 64 + 32 + 32 + 16 = 144$. Geometrically, the big square of side $(a+b)$ splits into four rectangles: a^2 , ab , ab and b^2 .



Area model for $a = 8$, $b = 4$ (rectangles drawn to scale).

Where mathematics like this is useful

- **Cryptography** – cryptographic algorithms use mathematical operations extensively; algebraic identities appear in the maths behind some systems.
- **Modular arithmetic** – security algorithms often calculate expressions under a modulus.
- **Algorithm development** – formulas help analyse the time and space complexity of security algorithms.
- **Security research** – algebra and number theory are the foundation for understanding encryption and protocols.
- **Programming / problem solving** – basic algebra helps when implementing and analysing security algorithms.

Important: you normally won't use $(a+b)^2$ directly in a penetration test or an Nmap scan. It is part of the mathematical foundation behind cybersecurity and cryptography.

2. What is cryptography?

Cryptography is the technique of protecting information by converting readable data (**plaintext**) into an unreadable form (**ciphertext**) using mathematical algorithms and a **key**.

```
HELLO    → Encrypted data → X7#kP9
X7#kP9   → Decrypted data  → HELLO
```

Main purposes

- **Confidentiality** – keeps information secret.
- **Integrity** – ensures data has not been changed.
- **Authentication** – verifies who sent the data.
- **Non-repudiation** – prevents someone from denying that they performed an action.

Common examples

- **AES** – encrypts data.
- **RSA** – public-key cryptography.
- **SHA-256** – creates a cryptographic hash.
- **Digital signatures** – verify authenticity and integrity.

In one line: Cryptography = protecting information using mathematics, algorithms, and keys.

Simple example

Without cryptography, a password sent as HELLO123 can be read by anyone who intercepts it. With encryption the flow is:

```
Original data (Plaintext)
      ↓
    Encryption
      ↓
Encrypted data (Ciphertext)
      ↓
    Decryption
      ↓
Original data

Plaintext:  HELLO
Key:        3
Encryption: Shift each letter by 3
Ciphertext: KHOOR
```

The receiver uses the correct key to convert KHOOR → HELLO. This is the basic idea behind the cryptography used to protect data during communication (a Caesar cipher).

3. CIA triad - main security goals

Goal	Meaning	Example
Confidentiality	Prevent unauthorized access to information.	Passwords and encryption.
Integrity	Prevent unauthorized modification of information.	Hashes to detect whether a file changed.
Availability	Ensure systems and information are available when needed.	Backups; protection against failures.

Easy to remember: C = keep it secret · I = keep it correct · A = keep it available.

Technical definitions

- **Confidentiality:** the security property that ensures information is accessible only to authorized entities and protected from unauthorized disclosure.
- **Integrity:** the security property that ensures data remains accurate, complete, and unaltered except by authorized actions.
- **Availability:** the security property that ensures authorized users can access systems, services, and data when required.

CIA Triad = Confidentiality + Integrity + Availability

4. Lab: SSH key setup between two AWS EC2 instances

Screenshots from a PuTTY session on two Amazon Linux 2023 servers: **ip-172-31-8-108** (left, user **ki1**) and **ip-172-31-7-154** (right, user **ki2**).

Steps shown

- Create a user and set a password: `sudo useradd ki1`, `sudo passwd ki1` (and **ki2** on the second server). The 'BAD PASSWORD: shorter than 8 characters' warning appears, but the password is still accepted.
- Edit `/etc/sudoers` and `/etc/ssh/sshd_config` with `sudo vim`.
- Generate an RSA 3072-bit key pair with `ssh-keygen` (a first attempt typed `ssh -keygen` failed with 'Bad escape character').
- Copy the public key to the other server with `ssh-copy-id user@65.1.95.75`; the host fingerprint was accepted with 'yes'.
- Result: Permission denied (publickey,gssapi-keyex,gssapi-with-mic) – the key was not installed, because the login used the generic name `user` and key/password login for that account was not enabled.

Likely fix: use a real account name (for example `ssh-copy-id ki1@65.1.95.75`) and make sure `PasswordAuthentication yes` is set in `sshd_config` (then `sudo systemctl restart sshd`) so `ssh-copy-id` can authenticate once with the password.

Screenshots

```
ec2-user@ip-172-31-8-108:~$ login as: ec2-user
Authenticating with public key "ki1"
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-8-108 ~]$ sudo vim /etc/sudoers
[ec2-user@ip-172-31-8-108 ~]$ sudo vim /etc/ssh/sshd_config
[ec2-user@ip-172-31-8-108 ~]$ sudo useradd ki1
[ec2-user@ip-172-31-8-108 ~]$ sudo passwd ki1
Changing password for user ki1.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-8-108 ~]$ ssh-keygen
Bad escape character 'ygen'.
[ec2-user@ip-172-31-8-108 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase for "/home/ec2-user/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:yf2aExhjN9MswitUQWba9vYfBgS6UZ/vTFKXjLphFk ec2-user@ip-172-31-8-108.ap-south-1.compute
The key's randomart image is:
+---[RSA 3072]-----+
|
|  + o
| * . o o
| . = . o B o
| o + S B . * . E
| o * . . + . .
| . . + o * B . |
| . o . o *
| . o o B
|-----[SHA256]-----+
[ec2-user@ip-172-31-8-108 ~]$ ssh-copy-id ^C
[ec2-user@ip-172-31-8-108 ~]$

ec2-user@ip-172-31-7-154:~$ Last login: Wed Oct 7 06:46:58 2026 from 13.233.177.5
[ec2-user@ip-172-31-7-154 ~]$ sudo useradd ki2
[ec2-user@ip-172-31-7-154 ~]$ sudo passwd ki2
Changing password for user ki2.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-7-154 ~]$ sudo vim /etc/sudoers
[ec2-user@ip-172-31-7-154 ~]$ sudo vim /etc/ssh/sshd_config
[ec2-user@ip-172-31-7-154 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase for "/home/ec2-user/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:GR6BxV5mc5kzNOZO/up1JiUJnvMM41BqDvRMCqYoh8 ec2-user@ip-172-31-7-154.ap-south-1.compute
The key's randomart image is:
+---[RSA 3072]-----+
|
|  + o
| * . o o
| . = . o B o
| o + S B . * . E
| o * . . + . .
| . . + o * B . |
| . o . o *
| . o o B
|-----[SHA256]-----+
[ec2-user@ip-172-31-7-154 ~]$ ssh-copy-id user@65.1.95.75
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ec2-user/.ssh/id_rsa.pub"
The authenticity of host '65.1.95.75 (65.1.95.75)' can't be established.
ED25519 key fingerprint is SHA256:zs5yaadl+nFdk4S/zl0dHLYE9+BYD20xri0z79U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that
are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is
to install the new keys
user@65.1.95.75: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
```

Screenshot 1 – both terminals after `ssh-keygen` and the `ssh-copy-id` attempt (permission denied).

```
ec2-user@ip-172-31-8-108:~$ login as: ec2-user
Authenticating with public key "ki1"

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-8-108 ~]$ sudo vim /etc/sudoers
[ec2-user@ip-172-31-8-108 ~]$ sudo vim /etc/ssh/sshd_config
[ec2-user@ip-172-31-8-108 ~]$ sudo useradd ki1
[ec2-user@ip-172-31-8-108 ~]$ sudo passwd ki1
Changing password for user ki1.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-8-108 ~]$ ssh-keygen
Bad escape character 'ygen'.
[ec2-user@ip-172-31-8-108 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase for "/home/ec2-user/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:yf2aExhjN9MswituQWba9YfbgS6Uz/vTPKxKjLphFk ec2-user@ip-172-31-8-108.ap-south-1.compute-1.amazonaws.com
The key's randomart image is:
+-----[RSA 3072]-----+
|
|  + o
| * . o o
| . = o B o
| o + S B . E
| o * . . + .
| . . + o * B
| . o . o *
| o o o B
|-----[SHA256]-----+
[ec2-user@ip-172-31-8-108 ~]$ ssh -copy-id ^C
[ec2-user@ip-172-31-8-108 ~]$
```

```
ec2-user@ip-172-31-7-154:~$ login as: ec2-user
Authenticating with public key "ki2"

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Wed Oct 7 06:46:58 2026 from 13.233.177.5
[ec2-user@ip-172-31-7-154 ~]$ sudo useradd ki2
[ec2-user@ip-172-31-7-154 ~]$ sudo passwd ki2
Changing password for user ki2.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-7-154 ~]$ sudo vim /etc/sudoers
[ec2-user@ip-172-31-7-154 ~]$ sudo vim /etc/ssh/sshd_config
[ec2-user@ip-172-31-7-154 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase for "/home/ec2-user/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:GR6BxV9mc5kzNOZO/upIJiUjnvMMW4lBqDvKMcQyOh8 ec2-user@ip-172-31-7-154.ap-south-1.compute-1.amazonaws.com
The key's randomart image is:
+-----[RSA 3072]-----+
|
|  + o
| * . o o
| . = o B o
| o + S B . E
| o * . . + .
| . . + o * B
| . o . o *
| o o o B
|-----[SHA256]-----+
[ec2-user@ip-172-31-7-154 ~]$ ssh-copy-id user@65.1.95.75
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ec2-user/.ssh/id_rsa.pub"
The authenticity of host '65.1.95.75 (65.1.95.75)' can't be established.
```

Screenshot 2 – earlier state: ssh-copy-id has just asked to confirm the host authenticity.

```
ec2-user@ip-172-31-8-108:~$ login as: ec2-user
Authenticating with public key "ki1"

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-8-108 ~]$ sudo vim /etc/sudoers
[ec2-user@ip-172-31-8-108 ~]$ sudo vim /etc/ssh/sshd_config
[ec2-user@ip-172-31-8-108 ~]$ sudo useradd ki1
[ec2-user@ip-172-31-8-108 ~]$ sudo passwd ki1
Changing password for user ki1.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-8-108 ~]$ ssh-keygen
Bad escape character 'ygen'.
[ec2-user@ip-172-31-8-108 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase for "/home/ec2-user/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:yf2aExhjN9MswituQWba9YfbgS6Uz/vTPKxKjLphFk ec2-user@ip-172-31-8-108.ap-south-1.compute-1.amazonaws.com
The key's randomart image is:
+-----[RSA 3072]-----+
|
|  + o
| * . o o
| . = o B o
| o + S B . E
| o * . . + .
| . . + o * B
| . o . o *
| o o o B
|-----[SHA256]-----+
[ec2-user@ip-172-31-8-108 ~]$ ssh -copy-id ^C
[ec2-user@ip-172-31-8-108 ~]$
```

```
ec2-user@ip-172-31-7-154:~$ login as: ec2-user
Authenticating with public key "ki2"

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Wed Oct 7 06:46:58 2026 from 13.233.177.5
[ec2-user@ip-172-31-7-154 ~]$ sudo useradd ki2
[ec2-user@ip-172-31-7-154 ~]$ sudo passwd ki2
Changing password for user ki2.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-7-154 ~]$ sudo vim /etc/sudoers
[ec2-user@ip-172-31-7-154 ~]$ sudo vim /etc/ssh/sshd_config
[ec2-user@ip-172-31-7-154 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase for "/home/ec2-user/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:GR6BxV9mc5kzNOZO/upIJiUjnvMMW4lBqDvKMcQyOh8 ec2-user@ip-172-31-7-154.ap-south-1.compute-1.amazonaws.com
The key's randomart image is:
+-----[RSA 3072]-----+
|
|  + o
| * . o o
| . = o B o
| o + S B . E
| o * . . + .
| . . + o * B
| . o . o *
| o o o B
|-----[SHA256]-----+
[ec2-user@ip-172-31-7-154 ~]$ ssh-copy-id user@65.1.95.75
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ec2-user/.ssh/id_rsa.pub"
The authenticity of host '65.1.95.75 (65.1.95.75)' can't be established.
ED25519 key fingerprint is SHA256:zs5yaadl+nFdkS4S/zlI0dHLYB9+EYD20xriI0z79U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that
are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is
to install the new keys
user@65.1.95.75: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-172-31-7-154 ~]$
```

Screenshot 3 – same session, final state with the 'Permission denied' error.